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PROJECT REVIEW -2

TITLE : Intelligent Disaster Data Interoperability and Emergency Response Coordination Platform

BY TEAM

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UNDER THE GUIDANCE OF

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Member 1: Disaster Data Integration and Emergency Response Coordination System (23MID0061)

Member 2 : Intelligent Disaster Assessment and Resource Management System (23MID0233)

AUDIO FILE (DRIVE LINK):

https://drive.google.com/file/d/1r0uXF4hqCt4MMy5anSfDQZXhmAIrK4id/view?usp=drivesdk

23MID0061_Salva Soumya_project review 2

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Speaker 00:00
I'm Soumya. Our project title is Disaster Data Integration and Intelligent Disaster Assessment System. So the main idea of our project is to make disaster related information easier to manage and also understand especially during the emergency situations. Usually the disaster information will be stored in different formats in different systems. So checking all the information separately can take a lot of time.

Speaker 00:29
Our project brings this information together and present it through a single platform. So the unique feature of our project is that it connects two different modules. The first module we have is that it will be integrating the data by transforming and validating also by schema mapping. Then the final process, the data will be sended to the second module, which

Speaker 00:55
will be used to display the incident details, the shelter information, the resources availability, the analytics and the suggested response actions. So instead of simply just displaying the raw data, our website organizes into meaningful sections so users can

Speaker 01:15
understand the situation more easily. So this will be helping them to identify the important incidents, review the resource shortages and plan their next steps more effectively. So our project is mainly useful for disaster management authorities, emergency response teams, also for the shelter managers.

Speaker 01:34
So basically the workflow is that first the data will be uploaded by the member1's module. It will be processed and then integrate the data set. Then sends the processed file to members2 module through an API. So member2 backends receives the file and the integrated incident records become available on the website.

Speaker 01:58
So when user opens the dashboard they get an overview of the incident situation in incident assessment they can also view the details such as the incident location the disaster type and the affected population where the information is available the shelter managers can

Speaker 02:17
check the shelter occupancy and the capacity while also we have the resource teams who can view the available supplies and the reported shortages. Also the analytics page will be showing the users to understand the patterns in the incident records and recommendations will be providing accordingly.

Speaker 02:41
So that the actions can be taken on the available information. So the main benefit is that the users can access the information relevant to their work through one website instead of checking separate records manually.

Speaker 02:56
So the tech stack that we have used for our project is that we used React with white for the front end. So also for the back end, we have used Python fast API. Then for the API connection, we use HTTP APIs to connect the member one's data integration module

Speaker 03:16
with member 2's disaster assessment module. So the Pandas helps us to read and work with the tabular data and CSV files. So during development, we run the front-end and back-end on our local system for now. We haven't deployed it. We are going to deploy the

Speaker 03:35
our project through the cloud platforms like Vercel or Render. Then since the two modules now running on separate laptops, we are using the CloudFare tunnel to allow the member ones back in to communicate with the members to back in. So for the final conclusion, our project main strength is that the connection between data integration and disaster assessment, we are building a platform that

Speaker 04:04
organizes disaster information and makes it easier for users to review the incidents, understand the resource conditions and support emergency response planning. Thank you ma'am.

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